SOFTWARE COST ESTIMATION REPORT

Using COCOMO II (Post-Architecture Model)

Project: Rudraksha Computers – Institute Management System

Tech Stack: HTML,CSS ,JS ,Node+Express | Developer: 1 | Duration: 5 Months

1.Project Overview

Parameter	Details
Project Name	Rudraksha Computers – Institute Management System
Tech Stack	HTML, CSS, JavaScript, Node.js, Express.js, MongoDB
Team Size	1 Developer
Development Duration	5 Months (December 2025 – April 2026)
Total SLOC (measured)	~14,000 Lines (Custom Code: Client + Server)
KSLOC	14.0 KSLOC
COCOMO II Model	Post-Architecture Model
Modules Count	12 Modules (Owner- Dashboard, Students, Courses, Packages, Attendance, Report Generation, Student-Dashboard, My courses (Download notes, Certificate Generation), Attendance, assignment)

2. COCOMO II Calculation Results

Parameter	Value
Model Formula	Effort = A × (KSLOC)^B × EM
Constants	A = 2.94, B = 1.09, EM = 0.78
Estimated Effort (COCOMO II)	45 Person-Months
Estimated Duration (COCOMO II)	12 Months (TDEV = 3.67 × Effort^0.317)
Actual Effort (Observed)	5.0 Person-Months (1 Developer × 4 Months)
Actual Duration (Observed)	5 Months
Number of Developers	1 (Solo Developer)
Billing Rate	₹12,000 per Person-Month
Development Cost (Actual PM × Rate)	**₹60,000 (5 PM × ₹12,000)**

3. Final Cost Analysis (Rudraksha Computers- Institute Management System)

Cost Category	Description	Estimated Cost (₹)
Development Cost	COCOMO II — 5 PM × ₹12,000/month (1 developer, 5 months)	60,000
Domain Name	Website domain (.com) — 1 year	800
Hosting (Frontend – Vercel)	Free / basic plan	0
Backend Hosting (Render)	Basic cloud hosting	2,000
Database (MongoDB Atlas)	Free tier / minimal usage	0
Internet & Electricity	Development + testing usage	2,000
Software & Tools	VS Code, Postman (Free)	0
Maintenance Cost (1 Year)	Minor updates, bug fixes	8,000
Miscellaneous Costs	Testing, setup, small expenses	2,000
Total		**74,800**